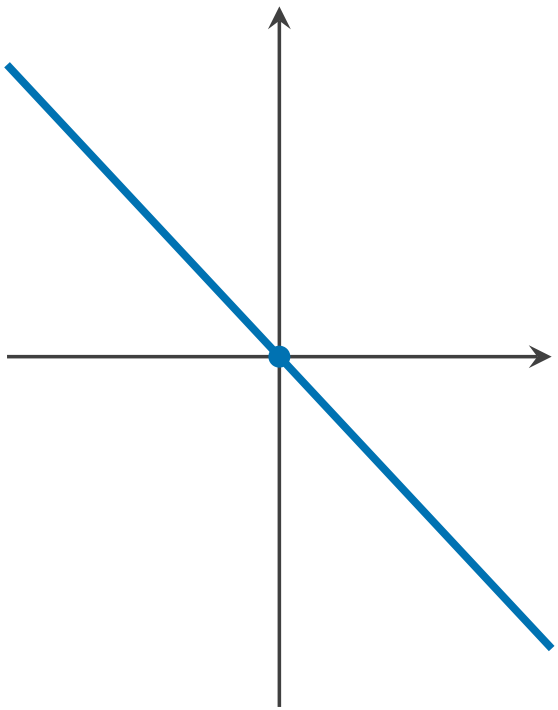
A. Regular crossing

$f(r) = -r$

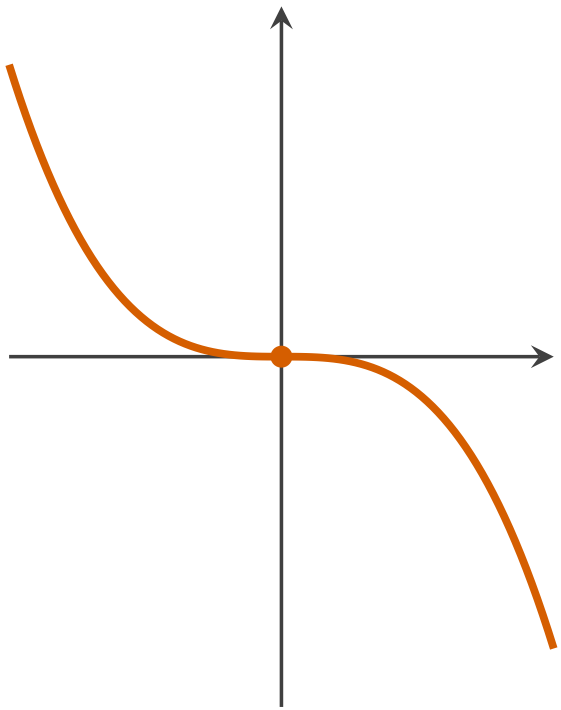


$f'(0) = -1 \neq 0$

$r = q - q^*$

B. Singular crossing

$f(r) = -r^3$



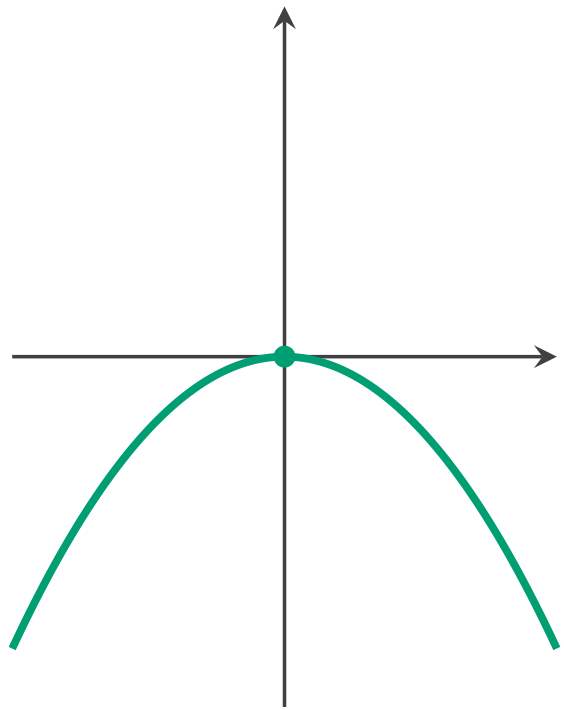
Unique zero, but

$f'(0) = 0$

$r = q - q^*$

C. Singular tangency

$f(r) = -r^2$



Vertical shifts

change root count

$r = q - q^*$